

Trainings ▾

Preparatory Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Do not bleed the fuel system of a hot engine; this can result in fuel spilling onto a hot exhaust manifold, which can cause a fire.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

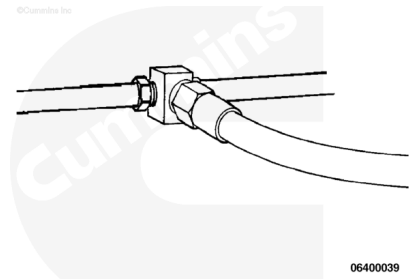
- Clean debris from fittings. Refer to Procedure 000-009 in Section (</qs3/pubsys2/xml/en/procedures/99/99-000-009.html>)
- Remove the air intake manifold. Refer to Procedure 010-023 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-023-tr.html>)

Remove

with Mechanically Actuated Injector

⚠ CAUTION ⚠

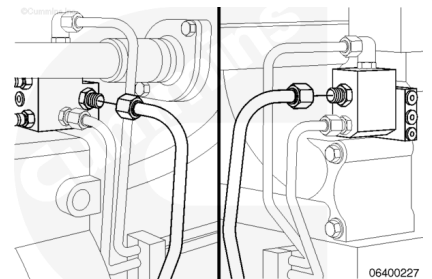
Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.



Remove the fuel drain hose that connects the engine fuel drain line to the fuel tank.

Disconnect the fuel drain line from the front of the fuel manifold.

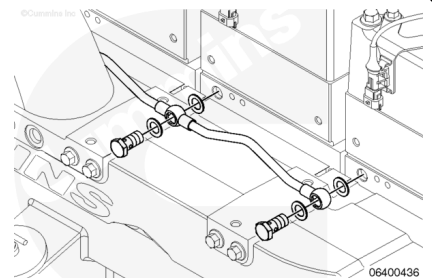
Disconnect the fuel drain line from the rear of the fuel manifold.



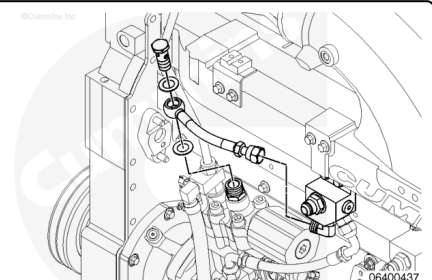
with Electronically Actuated Injector

Remove the bolt clamping the flexible drain line to the engine.

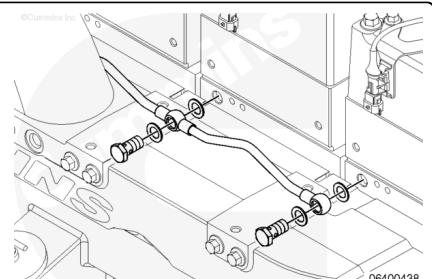
Remove the banjo capscrew securing the flexible line to the drain manifold.



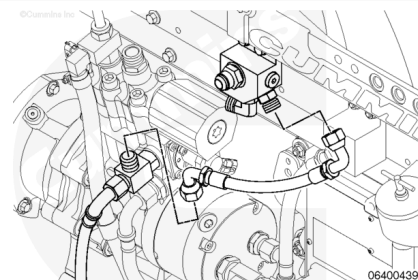
Remove the banjo capscrews and sealing washers securing the drain line between the mechanical dump valve and the drain manifold.



Remove the five remaining banjo capscrews, sealing washers, and the fuel drain line.



Remove the banjo capscrews and sealing washers securing the drain line between the injector supply line block and the drain manifold.

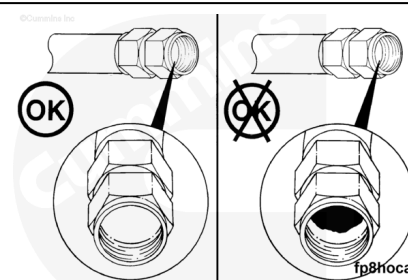


Inspect for Reuse

with Mechanically Actuated Injector

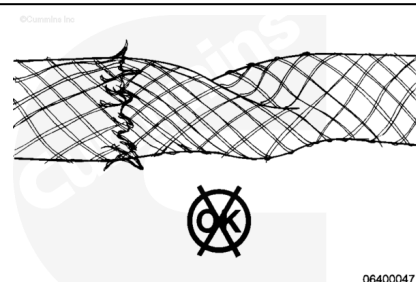
Inspect the inside of the fuel drain hose from the engine to the fuel tank for the following:

- The inner lining of the hose can separate from the center hose section
- A separation or flap can cause a restriction in the fuel flow.



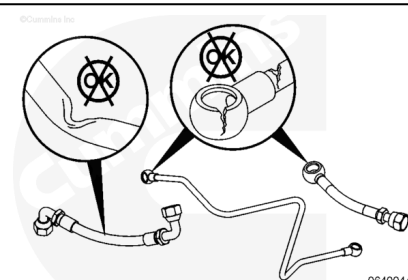
Inspect for any pinches in the hose that would obstruct the flow.

Inspect the fuel drain line for cracks and sharp bends.



with Electronically Actuated Injector

Inspect the fuel drain lines for cracks and other visible damage. The drain line **must** be replaced if it is cracked or otherwise damaged.

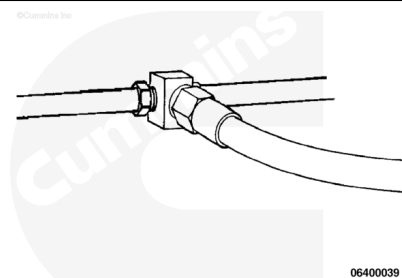


Install

with Mechanically Actuated Injector

Install and tighten the drain hose that connects the fuel drain line from engine to the fuel tank.

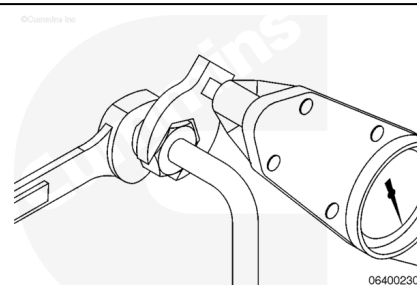
Torque Value: 75 n•m [55 ft-lb]



Connect the fuel drain line to the front fuel drain manifold.

Connect the fuel drain line to the rear fuel drain manifold.

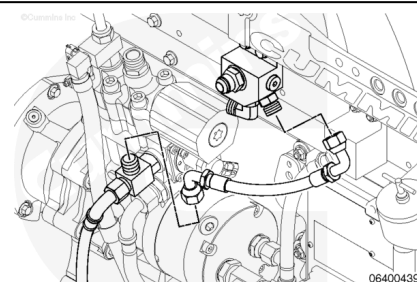
Torque Value: 48 n•m [35 ft-lb]



with Electronically Actuated Injector

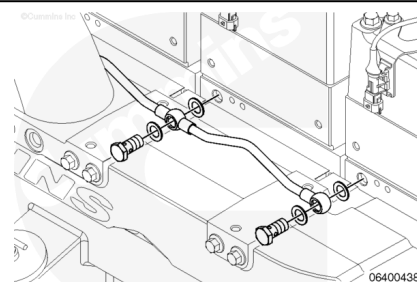
Install the banjo capscrews and sealing washers securing the drain line between the injector supply line block and the drain manifold.

Torque Value: 48 n•m [35 ft-lb]



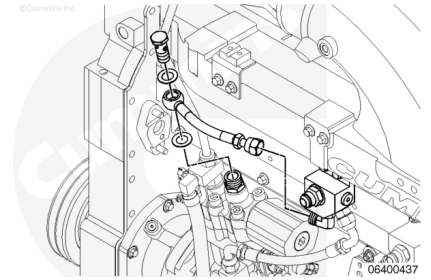
Install the fuel drain line, sealing washers, and the five banjo capscrews.

Torque Value: 22 n•m [195 in-lb]



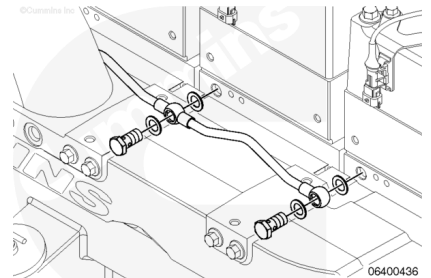
Install the banjo capscrews and sealing washers securing the drain line between the mechanical dump valve and the drain manifold.

Torque Value: 48 n•m [35 ft-lb]



Install the bolt clamping the flexible drain line to the engine.
Install the banjo capscrews securing the flexible line to the
drain manifold.

Torque Value: 22 n•m [195 in-lb]



Finishing Steps

with Electronically Actuated Injector

- Install the intake air manifold. Refer to Procedure 010-023 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-023-tr.html>)
- Operate the engine and check for leaks.

